

Théo Charlot

Curriculum Vitae

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Paris



Research interests : Human inspired machine learning, Developmentally-plausible machine learning, Human reasoning

Publications / Preprints

- 2026 **Charlot**, Kunze, Sheth, Cristia, and Lavechin, “Context-aware child-directed speech detection from long-form recordings”, **Interspeech**
- 2026 **Charlot**, Kunze, Poli, Cristia, Dupoux, and Lavechin, “BabyHuBERT: Multilingual Self-Supervised Learning for Segmenting Speakers in Child-Centered Long-Form Recordings”, **Interspeech**
- 2026 Mohapatra, **Charlot**, Duca, Palan, Romary, and Cassell, “Frame of Reference: Addressing the Challenges of Common Ground Representation in Situational Dialogs”, **Findings of ACL**

Professional Experiences

- Nov - Apr 2026 **Research Engineer @ École Normale Supérieure (ENS Ulm), CoML & LAAC teams**,
Who speaks to whom? Finetuning BabyHuBERT for child-directed speech (CDS) classification of adult speech.
- Apr - Aug 2025 **M2 : Research Intern @ École Normale Supérieure (ENS Ulm), CoML team**,
Who speaks when ? Pretraining and fine-tuning BabyHuBERT, a speech representation model trained on 13k hours of multilingual child-centered egocentric audios for segmenting speakers. → **VTC2.0** github repo
- Jun - Aug 2024 **M1 : Research Intern @ INRIA Paris, ALMAnaCH team**
Evaluation of LLMs ability to accurately retrieve precise information in long dialogs. Experiments with **inference** approaches to tackle context limit (**RAG**, **ChainOfThought**, **InfiniTransformers**, **MEMGpt**)
- Feb - Aug 2023 **Software Engineer @ TNS Mars**
In charge of implementing a lightweight AI assisted video annotation tool using openCV.

Diplomas

- 2023-2025 Master Degree in Computer Science : Apprentissage et Traitement Automatique de la Langue (ATAL)**, *Nantes Université*, *Highest Honors(rank: 1st/14)*.
Domains M2 : Language Modeling, Advanced NLP, Deep Learning, Scientific Communication
M1 : NLP, LLMs, LLM fine-tuning, Unsupervised Learning, Metaheuristics, Data Science, Algorithmics, Big Data
- 2018-2022 Bachelor in Computer Science**, *Sorbonne Université*, Paris.
Domains Reinforcement Learning, Supervised Learning, Data Science, Algorithmics, Web Fullstack, Linux Systems

Academic Projects

Summarization of U.S. Supreme Court Opinions (2024-25) - NLP Research Project

- Description Frugal on-device summarization for very long documents (100k+ tokens) with quantized models (Llama 1B Q4_K_M) and creation of an audio dataset (2k+ hours) for audio only arguments opinions

CEM-ERL (2022) - Research Project on Reinforcement Learning

- Description Developement and analysis of our own deepRL algorithm combining a SOTA population and gradient based algorithm (cem-rl) with evolutionary methods (ERL)

Winter Schools - Conference Workshops

- Apr - 2025 **ALPS2025, Advanced Language Processing winter School**.
Advanced NLP; Project sessions on reasoning traces; lectures on text generation, language modeling, agents
- May - Jul 2024 **DEFT2024 @ TALN2024, ExBERT**, *Travel grant from ATALA association*.
Mixture of Experts model fine tuned to answer medical multiple choice questions composed of 3 existing CamemBERT models pre-trained on medical data as experts following *Branch Train Mix* (Sukhbaatar et al. 2024)

Other

- Languages **French**: native, **Italian**: bilingual, **English**: TOEIC : 945/990, C1
- Associations Former Member of: **ATALA**; **Petits pieds nantais**(environmental); Nantes Université’s young researchers **LOGIN**